

Assigned: February 15, 2025

Due: February 28, 2025 (before 11.59 PM)

Marks: 20 marks possible

Submission Procedure:

- **Create / debug / test your assignment in Visual Studio**
- Create a **ZIP** file of your entire CareerCloud solution and upload it to **BlackBoard**.
 - o Please ensure you upload the entire solution
 - o Name your ZIP file according to the guidelines in the '1 – Sundry' slide deck
 - E.g. NET.31-6-Joe-Smith-N#####.zip
- **DO NOT EMAIL YOUR ASSIGNMENT** – The Humber email server will not deliver the assignment and you will not be notified that it was not delivered.

Description:

In this assignment you will create a gRPC service interface into the backend system.

Steps:

1. Add a new project to the 'CareerCloud' solution.
 - a. The new project will be of type 'gRPC Service' and **use .Net 8.**
 - b. The name of the new project will be 'CareerCloud.gRPC'
2. In the 'Protos' folder that was created as part of the 'CareerCloud.gRPC' project add a **protobuf definition file for each POCO mentioned below.**
3. In the 'Services' folder that was created as part of the 'CareerCloud.gRPC' project **add a service class for each POCO being exposed (09 no's).**
 - a. For example the 'ApplicantEducationPoco' class will have a corresponding service class called 'ApplicantEducationService'.
 - b. All service classes will inherit from their corresponding gRPC class.
 - c. All service classes will create an instance of their corresponding logic layer in their constructor and persist that instance in a private backing field.
4. Each service class will have the **methods for CRUD operations for records** stored in the database of the corresponding POCO.
 - a. All service methods will access data **only** through the business logic layer. Do not call methods on the repository layers directly, only the methods on the logic layer.
 - b. All controller methods will return a protobuf result type corresponding to their poco or void if the logic layer returns void.

POCOs to be implemented:

ApplicantEducationPoco.cs

ApplicantJobApplicationPoco.cs
ApplicantProfilePoco.cs
CompanyDescriptionPoco.cs
CompanyJobEducationPoco.cs
CompanyJobPoco.cs
SecurityLoginPoco.cs
SecurityLoginsLogPoco.cs
SystemLanguageCodePoco.cs

Marking Scheme:

Item	Maximum Mark	Your Mark
Add the CareerCloud.gRPC project	2	2
Protobuf contract files	9	9
Service classes	9	9
Total	20	20

Reference

Item	Reference
Protobuf scalar data types	https://docs.microsoft.com/en-us/dotnet/architecture/grpc-for-wcf-developers/protobuf-data-types
Google's C# gRPC programming guide	https://developers.google.com/protocol-buffers/docs/csharptutorial